

Harmony_FR_AI_003_Live_Translation_HLD

Item	Value or description
Feature ID	Harmony_FR_AI_003
Feature Name	Live Translation
Priority	P0
Type	AI / voice / Qira
Covered Layer	Pin
Status	High Level Design
Version	v1.0.1

Revision

Ver	Date	Author	Description
v1.0.0	2026/6/11	yuankm3	Live translation initial version for review
v1.0.1	2026/6/29	yuankm3	Clean up

Definition

- AIW-- AI wearable
- Auto stop-- it detects if current dialogue ended w/o explicit control by the user
- ASR-- Automatic Speech Recognition, it turns speech to text, so AKA STT
- LT-- Live Translation
- Source language-- the (mother or second) language of the AIW user speaks
- Target language- one of 10 languages the foreigner/s speak before or beside the AIW user
- TTS-- Text To Speech, it synthesize text to speech
- VAD-- Voice Activation Detection, it reports when someone speaks

Use case

Please refer to [PRD](#), which targets an offline face2face or side by side dual way translation using Harmony with a phone & buds being paired.

Note that Harmony doesn't have a display integrated, so a paired phone shall be present for dual way translation usage. If or not a single way translation shall be customized for cases like listening to a lecture in foreign language is TBD.

The UX when Harmony w/o buds paired is TBD.

High Level Design

How to trigger the live translation function is not discussed here.

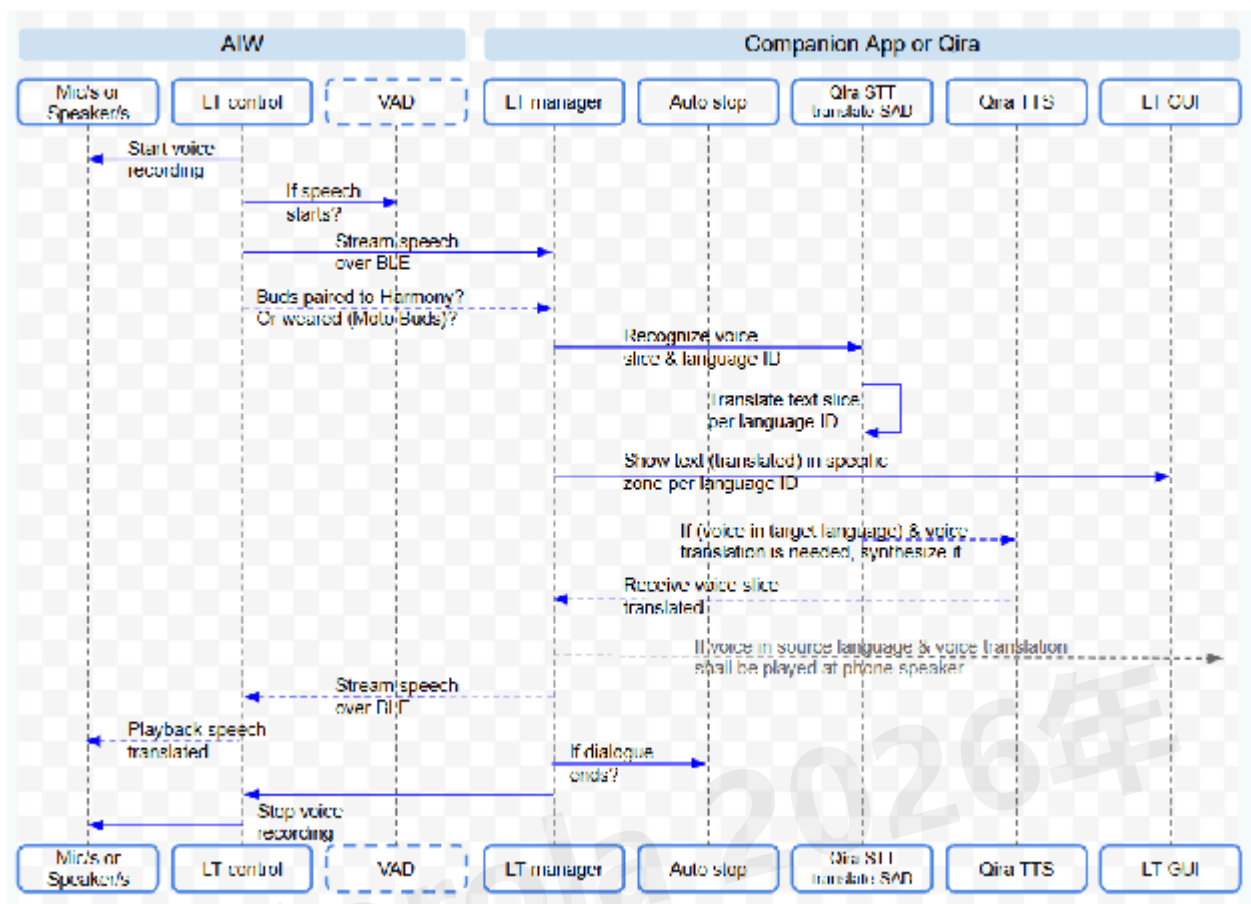
How to set source/target language is not discussed here.

How to change the layout (face2face or side by side) of the live translation GUI on the phone side is not discussed here.

How to change the voice recording mode on AIW per e.g. the GUI layout is not discussed here.

It's supposed that Qira STT translation can tell the language ID of a voice slice.

It's supposed that Qira STT translation includes the end of speech detection.



Key check points by SW Test

The latency from the user starts speaking in source language to the translated text starts displaying in CA: TBD

The latency from the foreigner starts speaking in target language to the translated voice starts playback in AIW: TBD